

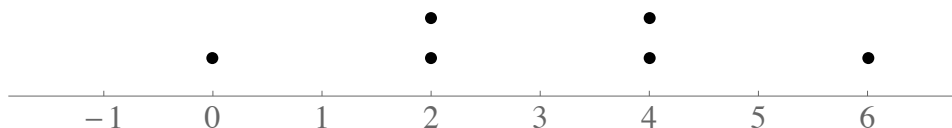
Chip-Firing Games—Solutions

Fall 2018 ARML Power Contest

Problem 1. Perhaps the easiest way to work this out is to proceed as follows. First, fire the 4-pile twice, leaving two chips in each of the 1-, 3-, and 5-piles.



Now, fire each of these piles, leaving the arrangement below (total firings: 5).



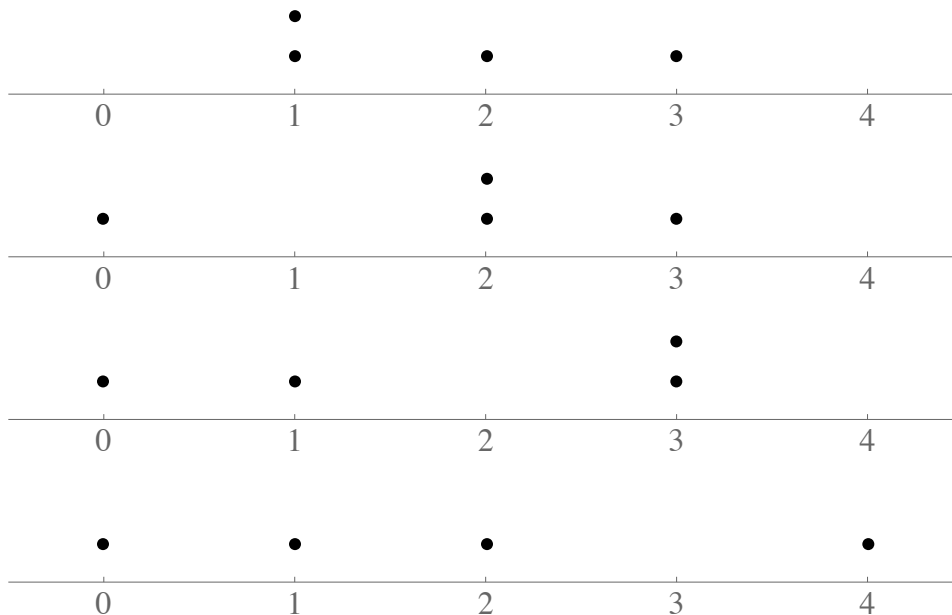
Now fire the 2-pile and the 4-pile (total firings: 7) to obtain the configuration below.



Finally, fire the 3-pile. Thus, after 8 firings a configuration is reached in which each pile has at most one chip and no more firings are possible. The final configuration is shown below.



Problem 2. The only pile that can be fired to start is the 1-pile, so fire it. This puts a chip on the 0-pile, and an additional chip on the 2-pile. The game is complete if $k = 1$, but if $k > 1$ there are now two chips on the 2-pile. So fire it, putting a chip on the 1-pile and an additional chip on the 3-pile. And so on. After piles 1 through a , with $a < k$, have been fired, there will be single chips on piles 0 through $a - 1$, no chips on a , two chips on $a + 1$, and one chip on piles $a + 2$ through k . Eventually there is one chip on piles 0 through $k - 2$ and two chips on pile k , with nothing beyond that. Finally firing the k -pile puts a single chip on piles 0 through $k - 1$ and on pile $k + 1$ with no other non-empty piles. The sequence is shown below for $k = 3$.



Note that each pile from 1 to k has been fired exactly once, so the procedure requires $\boxed{k}$ firings. Also note that the pattern described works when $k = 1$ as well, for then the starting configuration is just two chips in the 1-pile, which is fired to obtain the stable configuration with one chip in each of the 0-pile and the 2-pile.

Problem 3. If there is just one chip, then the game is already in a stable configuration so for $n = 1$ the final configuration is one chip in the 0-pile.

For $n = 2$, fire the 0-pile, leaving one chip each on the 1-pile and the -1 -pile. This configuration is stable.

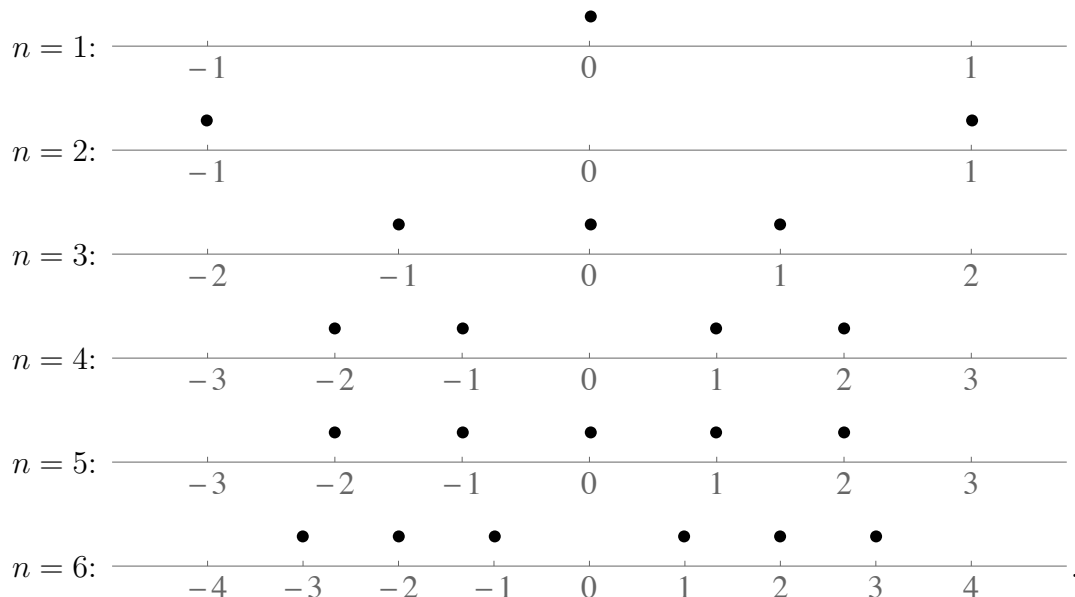
For $n = 3$, fire the 0-pile, leaving one chip each in the 1-, 0-, and -1 -piles, which is a stable configuration.

For $n = 4$, start by firing the 0-pile twice, putting two chips each in the 1-pile and the -1 -pile. Firing these puts one chip in the 2-pile and one in the -2 -pile, and two chips in the 0-pile. The 0-pile is now fired resulting in a stable configuration with one chip each on the -2 -, -1 -, 1-, and 2-piles.

The $n = 5$ case plays out just like the $n = 4$ case except the bottom chip on the 0-pile never moves, resulting in final configuration of one chip in each pile from -2 to 2.

Finally, for $n = 6$ note that we can play this out by spreading out the top four chips in the 0-pile as in the $n = 4$ case. This puts a single chip on piles -2 , -1 , 1 and 2 and leaves two chips in the 0-pile. Now fire the 0-pile. This leaves a situation like problem 2 with $k = 2$ on the positively numbered piles and a mirror image on the negatively numbered piles. Play each of these out, which will leave one chip on piles -3 , -1 , 1, and 3, and two chips on the 0-pile. Again fire the 0-pile, and then the 1- and -1 -piles, and finally the 0-pile again. We find that the final configuration is a single chip on each pile from -3 to 3 except that the 0-pile is empty.

Graphically, the answers are as shown below:



Problem 4. The previous solution more-or-less counted the number of firings as it went. There were no firings for $n = 1$ and only one for $n = 2$ and $n = 3$. For $n = 4$ and $n = 5$ we fired the 0-pile twice, then each of the 1- and -1 -piles once, then the 0-pile again, for a total of five firings.

For $n = 6$ it takes five firings to spread out the top four chips. Then one firing of the 0-pile to obtain the situation of problem 2 and its mirror image. Each of these requires two firings (total so far: 10) to reach the configuration with one chip on the -3 -, -1 -, 1 -, and 3 -piles and two in the 0-pile. We fire the 0-pile again (note that this puts two chips in each of the 1 - and -1 -piles, which is again the position of problem 2, this time with $k = 1$ —recursion, anyone?). We now fire piles 1, -1 , and finally 0 again. Total: 14 firings.

To summarize, the number of firings required for $n = 1$ through 6 are $0, 1, 1, 5, 5$ and 14 .

Problem 5. If n is even, the stable configuration resulting from starting with n chips in the 0-pile is a single chip in each pile from $-n/2$ to $n/2$ except the 0-pile, which is empty. If n is odd, the stable configuration is one chip in each pile from $-(n-1)/2$ to $(n-1)/2$.

It will be simpler to handle the even and odd cases of n separately, even cases first. So let $n = 2r$. Note that if $r = 1$ ($n = 2$) the stable configuration is a single chip in piles -1 and 1 . This will be the base case of an induction.

Now assume that the the stable configuration starting with $2(r-1)$ chips in the 0-pile is a single chip in each pile from $-(r-1)$ to $r-1$ except the 0-pile.

Starting with $2r$ chips in the 0-pile, we may think of them as 2 chips and another $2(r-1)$ chips. Leaving the bottom two chips alone, we inductively spread the top $2(r-1)$ chips (note that any firing available in the $2(r-1)$ -chip configuration is still available playing out the $2r$ -chip game, because all piles are the same except the 0-pile which has two extra chips in it). So we can move to a position where there is a single chip in each pile from $-(r-1)$ to $(r-1)$ and *two* chips in the 0-pile. Now fire the 0-pile.

The result of this is to put two chips in the 1 - and -1 -piles, and one chip in each other

pile from $-(r-1)$ to $(r-1)$ except that the 0-pile is empty. Thus, piles 1 through $(r-1)$ are in the situation of problem 2, and the mirror image of this situation obtains in the negatively-numbered piles.

According to problem 2, the result of playing this out is to end with a single chip in piles 0 through $r-2$, and one more chip in pile r . Of course, the negatively-numbered piles play out the same way, putting a single chip in piles 0 through $-(r-2)$ and one in pile $-r$. Since both the positive and negative put chips back into the 0-pile, there are again two chips in that pile. That is, the configuration at this point is a single chip in piles $-r, -(r-2), -(r-3), \dots, -2, -1, 1, 2, \dots, r-3, r-2, r$ and two chips in the 0-pile. Fire the 0-pile again.

The result of this is to once again put us in the position of problem 2, only this time with $k = r-2$. So we again play out that scenario, this time finishing with a chip in pile $r-1$ but no chip in pile $r-2$ (and similarly for piles $-(r-1)$ and $-(r-2)$, and two chips in the 0-pile. Firing the 0-pile again will put us in the position of problem 2, with $k = r-3$ (and its mirror image in the negative numbers, of course).

So we may continue using the result of problem 2 to fill in each pile from r down to 2 (technically, requiring another induction!). The final firing of the 0-pile puts chips on the 1- and -1 -piles, completing the proof in the even case.

Now that we have completed the proof in the case of $n = 2r$ even, assume $n = 2r + 1$. Leaving the bottom chip alone for the time being, the top $2r$ chips play out as in the even case to spread the chips to one chip in each pile from $-r$ to r except the 0-pile. But in actuality there is the bottom chip left in the 0-pile, so the proof is complete—there is a single chip in each pile from $-r$ to r .

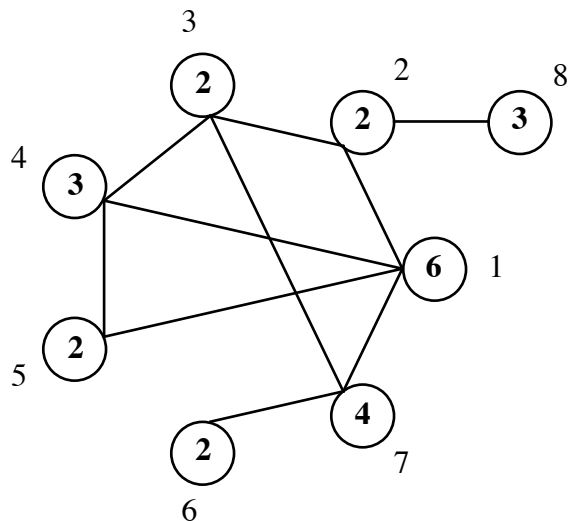
Problem 6. The inductive proof above shows how to count the firings needed to arrive at the stable configuration. First, note that from the last paragraph of the previous proof, the number of firings for the $2r + 1$ case is exactly the same as for the $2r$ case, so let $r = \lfloor n/2 \rfloor$. The total number of firings necessary is $\boxed{1^2 + 2^2 + \dots + r^2 = r(r+1)(2r+1)/6}$.

This figure can be arrived at as follows. Let $F(r)$ be the number of firings needed when $n = 2r$. Then, as in the induction above, after $F(r-1)$ moves there are two chips in the 0-pile and one in each other pile from $-(r-1)$ to $r-1$.

At this point, we fired the 0-pile, leaving the position of problem 2 with $k = r-1$ in both the positive and negative sides. Each of these requires $r-1$ firings, so we have performed $1 + 2(r-1) = 2r-1$ firings. And now the 0-pile fires again, returning to the scenario of problem 2 but with $k = r-2$. And so forth. So to complete the entire induction step requires $(2r-1) + (2(r-1)-1) + \dots + 3 + 1$ firings. It is well known that the sum of the odd number up to $2r-1$ is r^2 . Thus, we see that the difference $F(r) - F(r-1) = r^2$. Since when $r = 1$ it requires $1^2 = 1$ firing, the given formula holds for all r .

Problem 7. The result of firing a vertex is that it loses a number of chips equal to its valence, while each connected vertex gains one chip. So when all the initially fireable vertices P_4 through P_8 are fired the result is that P_1 gains three chips since it can't fire but has three neighbors that can, so it finishes with six chips. On the other hand, P_4 loses three but only gains one, finishing with three chips. The other vertices can be handled similarly. Of course,

none of the connections between the piles will change, so the final result will be as shown in the diagram.



Problem 8. First, firing P_x can only add chips to P_y , not take any chips away. So P_y can be fired after P_x as well as before it, and vice versa. Second, each pile other than P_x or P_y gains two chips if it is connected to both P_x and P_y , regardless of the order in which they are fired. Similarly, a pile that is connected to only one of P_x or P_y will gain one chip regardless of order, while a pile that is connected to neither will not change. Under both firing orders, both P_x and P_y lose a number of chips equal to their valence, both gaining one back if they are connected to each other. So all piles end up with the same number of chips regardless of the order in which the firings are performed.

Problem 9.

(a) If a pile has at least as many chips as its valence, then it can fire. So the only way the position can be stable is if each pile has at most $v_P - 1$ chips, where v_P means the valence of pile P . Adding this over all the piles gives $\sum(v_P - 1) = \sum v_P - P$. But note that the sum of all the valences of all the piles must equal $2C$ because each connection adds one to the valence of two different piles. Thus, a stable configuration has at most $2C - P$ chips, and more than this number of chips will always have at least one pile with at least as many chips as its valence by the pigeonhole principle.

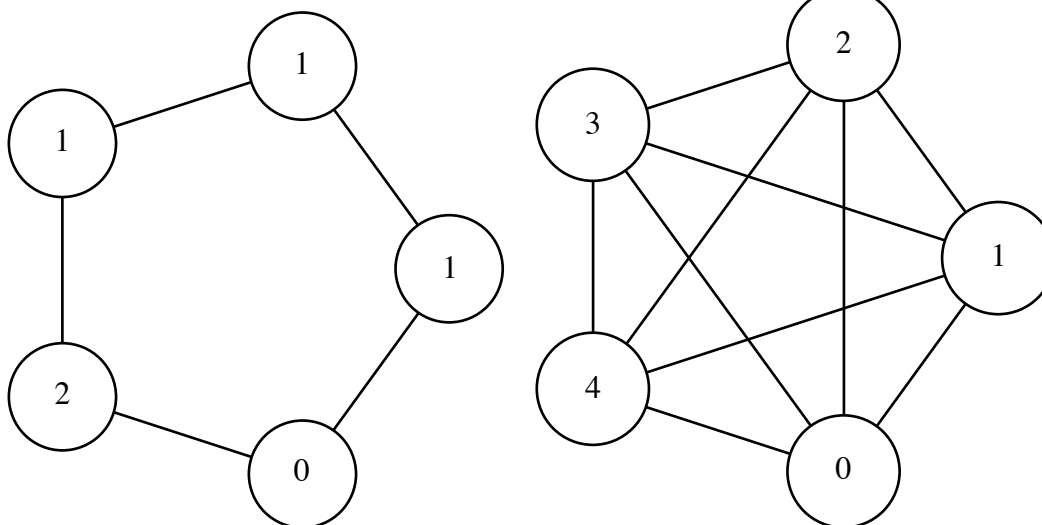
(b) If each pile has exactly $v_P - 1$ chips, no pile can fire and the configuration is stable. As noted above, this gives a total of $2C - P$ chips.

Problem 10. By the pigeonhole principle there must be at least one pile P_a that fires infinitely often. Choose any pile to which it is connected, P_b . Every time P_a fires, P_b gains a chip. Of course, P_b can never have more than N chips, as that is the total number of chips in the game. So P_b must fire infinitely often to avoid acquiring more than N chips. So any neighbor of a pile that fires infinitely often must itself fire infinitely often.

Since the game board is connected, every pile is either connected to P_a , or connected to a pile that is connected to P_a , or Every pile is connected to a pile that fires infinitely often, and therefore fires infinitely often itself.

Problem 11. Assume that the game reaches a stable configuration but that every pile has fired at least once. Consider the pile P_r that has fired least recently. Since all piles have fired, certainly every pile to which P_r is connected has fired since the last time P_r fired. But then P_r has acquired at least one chip from each pile to which it is connected, and currently must contain at least as many chips as its valence, so it can fire and the game has not reached a stable configuration. This contradiction shows that if each pile fires at least once a stable configuration can never be reached, proving the original statement by contradiction.

Problem 12. The arrangement below left is one initial configuration that will never lead to a stable configuration. The justification is that firing the only pile that can be fired—the pile with two chips in it—leads to a configuration that is identical except “rotated” one position clockwise. So the new configuration also has a pile that is ready to fire, and the game will continue forever.



Problem 13. The arrangement above right is one initial configuration that will never lead to a stable configuration. The only pile that can fire is the one with four chips in it, and firing leads to an identical configuration “rotated” one position clockwise as in the previous solution.

Problem 14. Consider the pile which currently has no chips in it. If the current distribution was arrived at by firing a pile in another configuration, the no-chip pile must have been the pile that fired. This is because if any other pile fired, since each pile is connected to each other pile the zero-chip pile would have received a chip from that other pile, and couldn't have zero chips in it now! So the zero-chip pile was the pile that fired to achieve the current distribution. Since it now has zero chips and is connected to $n - 1$ other piles, it had $n - 1$

chips in it. Each other pile received one chip from it. That is, the piles that currently have one, two, $\dots$, $n-1$ chips used to have zero, one, $\dots$, $n-2$ chips. So the previous configuration also had one pile with zero chips, one with one, $\dots$, one with $n-1$ chips as required.

Problem 15. Since there are only a finite number of chips and finite number of piles, there are only a finite number of configurations. Since the game never becomes stable some configuration must repeat. Now between two repetitions of the same configuration *every* pile must fire at least once. Otherwise, we could repeat the sequence of firings that leads from the repeated configuration to itself over and over, giving an infinite game that never fires some particular vertex, contradicting the result of problem 10.

Now consider the firings between one pair of repeats of a configuration. For each pile, there is a last time that it fires before the repeated configuration shows up again. For a pile p , let f_p be the number of (other) piles that fire after this last firing but before the configuration repeats. For example, if from a certain configuration the piles fire in order 5, 3, 4, 5, 2, 1, 3, 2, 3 before the configuration is restored, then $f_1 = 3, f_2 = 1, f_3 = 0, f_4 = 6$, and $f_5 = 5$. Note that f_p is defined for each pile because each pile must fire between repetitions. Also note that for different piles p and q it must be the case that $f_p \neq f_q$; in fact if the last firing of p is before the last firing of q then $f_p > f_q$. Finally, note that when the configuration repeats, each pile has at least f_p chips in it because it has acquired one chip from each other pile that fired after it, and may have had chips remaining in it after it fired for the last time.

Combining these facts about the f_p it becomes clear that at the time of the repeat, there must be a pile with at least zero chips, a different pile with at least one chip, a different pile with at least two chips, and so on until one pile that has at least $n-1$ chips. Adding these, there are at least $0 + 1 + 2 + \dots + (n-1) = n(n-1)/2$ total chips in the game. But since the total number of chips is $n(n-1)/2$, no pile may have any extra chips in it at the time of the repeat. That is, at the time the configuration first repeats, there is a pile of zero chips, a pile of one chip, $\dots$, a pile of $n-1$ chips. As shown in problem 14, this distribution is a “garden of eden” and could not have come from any other distribution. Thus, the only distribution of $n(n-1)/2$ chips on a complete game board of n piles that never becomes stable is this one.